Question	Answer	Marks
6(a)	The magnitude of the induced e.m.f. is **directly proportional** to **the rate of change of magnetic flux linkage or (rate of cutting of magnetic flux).**	[1] [1]
6(b)(i)	$\omega = \left(180\frac{rev}{\min}\right)\left(\frac{2\pi\ rad}{rev}\right)\left(\frac{\min}{60\ s}\right) = (180)(2\pi)\left(\frac{1}{60}\right)\frac{rad}{s} = 18.8\ rad\ s^{-1}$	[1]
6(b)(ii)	1. Correct amplitude (8.00 x 10^{3} Wb turns) 2. Correct period, at least two cycles (T is 0.333 s) 3. Sine or negative sine graph	[1] [1] [1]
6(b)(iii)	1. Cosine or negative cosine depending on **6(a)(ii)** 2. Correct amplitude for current (1.51 mA).	[1] [1]
6(b)(iv)	Max e.m.f. = 0.302 V By Faraday's Law, $\varepsilon = -N\frac{d\Phi}{dt}$. When the rate of rotation is doubled, rate of change of flux $\frac{d\Phi}{dt}$ will also double. Hence e.m.f. generated will double as well.	[1] [1]
	Max Marks	**10**

Question	Answer	Marks
7(a)	Once the skateboarder is airborne, he is subjected to air resistance while the skateboard is	**[1]**